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THE DEAD SALMONS OF AI INTERPRETABILITY



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ABSTRACT

In a striking neuroscience study, the authors placed a dead salmon in an MRI scanner and showed it images of humans in social situations. Astonishingly, standard analyses of the time reported brain regions *predictive* of social emotions. The explanation, of course, was not supernatural cognition but a cautionary tale about misapplied statistical inference. In AI interpretability, reports of similar "dead salmon" artifacts abound: feature attribution, probing, sparse auto-encoding, and even causal analyses can produce plausible-looking explanations for randomly initialized neural networks. In this work, we examine this phenomenon and argue for a pragmatic **statistical–causal reframing**: explanations of computational systems should be treated as parameters of a (statistical) model, inferred from computational traces. This perspective goes beyond simply measuring statistical variability of explanations due to finite sampling of input data; interpretability methods become statistical estimators, and findings should be tested against explicit and meaningful **alternative computational hypotheses**, with uncertainty quantified with respect to the postulated statistical model. It also highlights important theoretical issues, such as the identifiability of common interpretability queries, which we argue is critical to understand the field's susceptibility to false discoveries, poor generalizability, and high variance. More broadly, situating interpretability within the standard toolkit of statistical inference opens promising avenues for future work aimed at turning AI interpretability into a pragmatic and rigorous science.

1 INTRODUCTION

In 2009, researchers placed a dead salmon in an MRI scanner, showed it photographs of humans in social situations, and ostensibly asked it to judge their emotions (Bennett et al., 2009). Standard analysis pipelines commonly used at the time surprisingly returned brain voxels as significantly predictive of emotional situations. The error arose from a failure to correct for multiple comparisons within the statistical analysis pipeline. The "dead salmon" demonstration of false positives contributed to a larger reckoning in the field of neuroscience. For instance, an influential study showed that different research groups obtained different results even when analyzing the same dataset and the same research question (Botvinik-Nezer et al., 2020). Subsequent work identified several sources of *statistical fragility*. Widely used statistical procedures embedded in standard analysis pipelines were shown to inflate false-positive rates (Eklund et al., 2016), an effect worsened by non-independent analyses producing spuriously large brain–behavior correlations (Vul et al., 2009). Also, early neuroimaging research was constrained by small samples and limited data availability (Button et al., 2013; Marek et al., 2022), exacerbating overfitting and spurious associations. Moreover, fMRI had been criticized for offering predictive explanations, rather than functional ones, resulting in little clinical relevance (Lyon, 2017). Finally, reverse inference emerged as a central interpretative problem, given that individual neural systems are not uniquely associated with specific cognitive functions (Poldrack, 2006; Duncan & Owen, 2000).

AI interpretability now faces its own *dead salmon* issues, similarly begging for a larger reevaluation of its statistical foundations. A growing body of work has shown that many influential methods, including feature attribution (Adebayo et al., 2018), probing classifiers (Ravichander et al., 2021),

sparse autoencoders (Heap et al., 2025), circuit discoveries (Méloux et al., 2025), and causal abstractions (Sutter et al., 2025), can yield plausible-looking explanations even when applied to **random neural networks**. In Figure 1, we report a minimum dead salmon artifact from analyzing activations of a fully randomized BERT model in a sentiment analysis task where both correlation analysis and probing find highly significant *explanations*. Such striking failure modes are particularly troubling as modern AI systems are increasingly deployed in high-stakes domains where AI interpretability should be essential for transparency, accountability, and error diagnosis (Mehrabi et al., 2021; Barnes & Hutson, 2024; Ramachandram et al., 2025). Interpretability methods have the potential surface critical failure modes (Kim & Canny, 2017; Zech et al., 2018; Caruana et al., 2015; Meng et al., 2022; Monea et al., 2024; Nguyen et al., 2025) and offer levers for mitigating bias and systematic errors (Arrieta et al., 2019; Kristofik, 2025; Lepori et al., 2025).

Yet, despite frequent analogies to mature sciences like *neuroscience* (Barrett et al., 2019), *biology* (Lindsey et al., 2025), or *physics* (Allen-Zhu & Li, 2023; Allen-Zhu, 2024) of neural networks, the practice of AI interpretability remains in its early foundational stages. Striking dead-salmon artifacts are accompanied by a general statistical fragility: small perturbations to inputs (Ghorbani et al., 2019; Kindermans et al., 2019; Zhang et al., 2025) or changes in random initialization (Adebayo et al., 2018; Zafar et al., 2021) can radically change explanations. Explanations often fail to generalize to new settings and input distributions (Hoelscher-Obermaier et al., 2023). Also, multiple incompatible explanations can be *discovered* for the same behavior (Méloux et al., 2025; Dombrowski et al., 2019). While the dead salmon study demonstrated a simple statistical oversight correctable through multiple comparison adjustments, AI interpretability's difficulties stem from more fundamental issues. In particular, we argue that, for common interpretability queries, computational traces do not uniquely determine explanations.

Beyond neuroscience and AI, such challenges are not unprecedented. Psychology and the social sciences faced a similar reckoning during the replication crisis, when questionable research practices produced widespread false positives (Collaboration, 2015; Simmons et al., 2011; Ioannidis, 2005; Schimmack, 2020). These fields responded with methodological reforms: pre-registration, registered reports, increased statistical power, and explicit multiple-comparison corrections (Munafò et al., 2017; Korbmacher et al., 2023). Likewise, econometrics used causal inference (Pearl, 2009) to formalize the distinction between correlation and causation, developing identification criteria, sensitivity analyses, and robustness tests (Imbens & Rubin, 2015; Angrist & Pischke, 2009; Heckman, 2007).

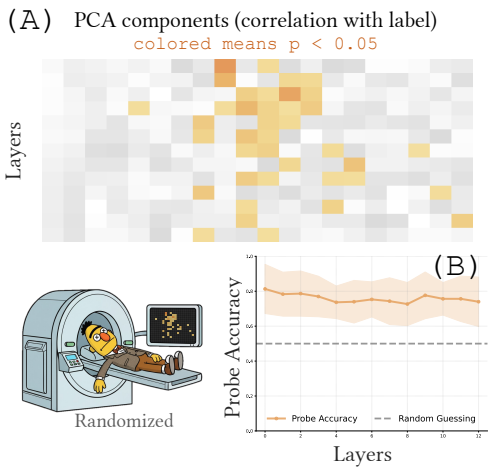


Figure 1: **Minimal dead salmon artifacts**. We extract token representations from a randomly initialized BERT for 300 IMDb sentences and average over sequence length. (A) Several principal components are spuriously correlated with sentiment labels. (B) A simple probe trained on layer representations achieves nontrivial cross-validated accuracy.

Now, AI interpretability can also begin to build its own methodological guardrails. As argued before, this requires both technical innovation and philosophical clarity (Miller, 2019; Williams et al., 2025). This means clarifying our epistemic goals by answering: what does it mean to "explain" a neural network? (Lillicrap & Kording, 2019; Lipton, 2018) Mechanistic interpretability embodies a type of *scientific realism*, aiming to discover the *one true* explanatory algorithm (Psillos, 2005; Chakravartty, 2011). However, there is a significant push-back against the feasibility of this research project (Rudin, 2019; Páez, 2019; Saphra & Wiegreffe, 2024), motivating a shift toward pragmatic approaches prioritizing the utility of the explanations for specific downstream goals (Zou et al., 2025). Here, we align with the pragmatic stance (Dewey, 1948; Chang, 2004; Potochnik, 2017), where explanations are seen as useful models that enable prediction, manipulation, and control (Van Fraassen, 1980; Cartwright, 1983).

This work. We analyze failure modes of contemporary AI interpretability methods, ranging from striking dead-salmon false positives to broader forms of statistical fragility, including poor generalization and high variance. We argue that these pathologies share a common root cause: the non-identifiability of many interpretability queries, compounded by the lack of principled uncertainty quantification, where non-identifiability manifests as high-variance estimates that should be reflected in large uncertainty. Diagnosing and addressing these issues, as well as articulating a coherent pragmatic research direction for interpretability, requires reframing AI interpretability as a problem of statistical (causal) inference. Accordingly, we propose one such statistical–causal reframing in which explanations are treated as parameters inferred from computational traces, enabling uncertainty-aware evaluation against meaningful alternative computational hypotheses.

2 The Statistical Fragility of AI Interpretability

Reports documenting the failure modes of interpretability methods are frequent and highlight a recurring theme: a general statistical fragility, most strikingly illustrated by dead salmon artifacts. We provide here a non-exhaustive overview of such issues.

Feature Attribution. Gradient-based attribution methods (Simonyan et al., 2014; Sundararajan et al., 2017) aim to highlight input features most relevant to model predictions. However, Adebayo et al. (2018) demonstrated that saliency maps can remain visually plausible even after model weights are randomized. Further, Dombrowski et al. (2019) showed that gradient-based explanations can be manipulated by adversarial perturbations, leaving predictions unchanged, while Ghorbani et al. (2019) revealed that explanations are unstable under minor data transformations. From a theoretical standpoint, Bilodeau et al. (2024) established impossibility results showing that no attribution method can simultaneously satisfy intuitive desiderata across broad model classes.

Probing. Probing methods train a classifier to predict a target label from internal activations. Early studies already showed that both linear and structural probes could recover information with surprisingly high accuracy from randomized contextualized embeddings (Conneau et al., 2018; Hewitt & Manning, 2019), and syntactic probes do not generalize (Hall Maudslay & Cotterell, 2021). Later, Ravichander et al. (2021) demonstrated that probes can extract features merely encoded (e.g., inherited from embeddings) even if unused during inference; probing asks whether a concept is encoded in an activation, not whether it is computationally relevant. Capacity-controlled and information-theoretic probes (Voita & Titov, 2020; Zhu & Rudzicz, 2020; Pimentel et al., 2020; Belinkov, 2022) or amnesic probing (Elazar et al., 2021) attempt to mitigate such false discoveries.

Sparse Autoencoders. Unsupervised concept-discovery pipelines such as sparse autoencoders (SAEs) (Cunningham et al., 2023; Yun et al., 2021; Bricken et al., 2023; Templeton et al., 2024) display analogous pathologies. Heap et al. (2025) showed that SAEs can recover apparently interpretable components even in randomly initialized transformers. Additional studies show that SAEs often fail to generalize across settings or tasks (Heindrich et al., 2025; Kantamneni et al., 2025). Also, Li et al. (2025a) show SAE are sensitive to adversarial input perturbations.

Concept-Based Explanations. Concept-based methods (Kim et al., 2018; Bau et al., 2017) aim to identify human-interpretable concepts that align with model representations (e.g., concept activation vectors (Kim et al., 2018) or network dissection (Bau et al., 2017)). These methods also face documented limitations (Sinha & Zhang, 2025; Aysel et al., 2025). Already, Bolukbasi et al. (2021) showed *interpretability illusion* arising where activations of individual neurons in BERT may spuriously appear to encode a concept. Then, Nicolson et al. (2025) showed that concept activation scores can produce highly inconsistent explanations, and Ramaswamy et al. (2023) documented poor generalization and high sensitivity to the dataset used to infer concepts. Finally, Piratla et al. (2024) further demonstrated high variance and recommended incorporating uncertainty estimation.

Causal Approaches. To address issues with prediction-based explanations, a shift toward causality-based interpretability has emerged through the use of causal mediation analysis (Pearl, 2012; Elazar et al., 2021; Vig et al., 2020b; Meng et al., 2022; Finlayson et al., 2021; Syed et al., 2023; Monea et al., 2024; Mueller et al., 2024). These methods intervene on intermediate representations to quantify causal effects of components on model outputs. Yet recent work documents substantial